Math 1232-20: Single-Variable Calculus II

GWU Summer 2025 Session II

Meetings:	MTWR, 4:00 - 5:30 1957 E Street Room 310 06/30/2025 - 08/09/2025	**Textbook:**	OpenStax Calculus Vol. 1 OpenStax Calculus Vol. 2 by Strang and Herman
Instructor:	Ben Clingenpeel	**Office:**	Phillips 720G
Email:	ben.clingenpeel@gwu.edu	**Office hours:**	1:30-3:00, MWF or by appointment
Course page:	https://bncling.github.io/courses/math-1232		

Course content and learning outcomes

Though condensed into just six weeks, this course covers the material of a second semester of a standard year-long sequence in single-variable calculus: the calculus of exponential and logarithmic functions, L'Hôpital's rule, techniques of integration, infinite series and Taylor series, and polar coordinates

By the end of the course, students will acquire the following skills and knowledge. Students will be able to:

- define logarithm, exponential, and inverse trigonometric functions, explain their basic properties (continuity, derivatives, asymptotes, etc.) and recognize their graphs;
- apply these functions to word problems, and correctly interpret the results;
- solve integrals using integration by parts, trigonometric substitution and partial fractions;
- determine whether an infinite series converges or diverges;
- express algebraic and transcendental functions using Maclaurin and Taylor series; and
- analyze, create and recognize polar and parametric graphs.

Textbook and additional Resources

Course textbook: Our primary text is OpenStax Calculus Volume 2 by Gilbert Strang and Edwin Herman. Some topics often covered in Calculus I at other institutions are deferred to Calculus II at GW, and so we will begin the course using our secondary textbook, OpenStax Calculus Volume 1, also by Gilbert Strang and Edwin Herman. Both texts can be read online for free at the OpenStax website. As a student in math classes, I have always found it helpful to read at least a little about a topic and try a problem or two before it is covered in lecture—it gives an idea of what to expect in the lecture and helps me think of some questions to ask in advance (and please do ask them). During each lecture I'll try to keep you oriented as to what part of the textbook we're covering, and at the end I'll suggest some textbook problems to work on.

Peer tutoring: GW offers a peer tutoring service through the Academic Commons. You can book a completely free 50 minute tutoring session here: https://academiccommons.gwu.edu/tutoring.

Mental health services: The University's Mental Health Services offers 24/7 assistance and referral to address students' personal, social, career, and study skills problems. Services for students include: crisis and emergency mental health consultations confidential assessment, counseling services (individual and small group), and referrals. For additional information see: https://counselingcenter.gwu.edu/ or call 202-994-5300.

Course structure

The course meeting times will be used for lectures, and although I won't be taking attendance, going to these lectures will be essential—we'll discuss the theory of calculus as well as numerous example problems similar to what you will see on homework and exams. Outside of the lectures, you will have regular homework, and in addition to the time you spend on this, you should also review your notes regularly (the guideline here is that you should try to spend 2 hours outside of class for every hour spent in class). Your grade in the course will be based on the following:

- Online homework through WeBWorK – 18%
- Mastery points – 32% (1pt = 1%)
- A midterm exam – 20%
- A final exam – 30%

Minimum scores for each letter grade are as follows: A, 93%; A-, 90%; B+, 87%; B, 83%; B-, 80%; C+, 77%; C, 73%; C-, 70%; D+, 67%; D, 63%; D-, 60%.

WeBWorK: Accessed through Blackboard under the **Assignments** tab, WeBWorK is an online homework system. The point of WeBWorK is to give you a good amount of problems to practice with. You never know how well you understand something until you try applying it, and as such you will get as many attempts as you need to answer all WeBWorK problems. There is, however, a due date for each problem set, usually a few days after the set becomes available. If you need more time, there is always a reduced scoring period available where you can still earn 90% credit up to four days after the set is due. Ideally, you should think of the reduced scoring period as a last resort. Staying on top of the WeBWorK is one of the best ways to help yourself through the course.

Mastery quizzes, mastery points: In addition to the WeBWorK, these will be the main homework. These quizzes cover 12 topics that are core to a second semester of calculus, and will be graded based on *mastery*. Each quiz will feature a few different topics, and your performance on each topic will be graded on a two-point scale, where a 0/2 means you've shown little-to-no understanding of the topic, and a 2/2 means you've demonstrated mastery of the topic. The topics are the following:

Major topics	Secondary topics
1. Transcendental Functions	1. Invertible Functions
	2. L'Hôpital's Rule
2. Integration Techniques	3. Improper Integrals
	4. Arc Length and Surface Area
3. Series Convergence	5. Differential Equations
	6. Sequences and Series
4. Taylor Series	7. Power Series
	8. Parameterization

You will have a chance to show mastery on each secondary topic twice, i.e. for any secondary topic, there will be two mastery quizzes with problems on that topic (among other topics). For a major topic, you will have four chances.

Every secondary topic is worth 2 mastery points, and every major topic worth 4. Your mastery score for any secondary topic is your best score on that topic, and your mastery score for any major topic is the sum of your best two scores on that topic. For example, if on the mastery quizzes for S3 (Improper Integrals) you get a 1/2 and then a 2/2, your mastery score for S3 is a 2/2. If for M3 (Series Convergence) you get a 0/2, a 1/2, another 1/2, and then a 2/2, your mastery score for M3 is a 3/4.

Your final mastery score is the sum of all of your mastery points. There are 8 secondary topics, worth 2 points a piece, and 4 major topics, each worth four points. Your final mastery score is therefore some number between 0 and 32, and this part of your grade is weighted so that every mastery point is worth 1% of your final grade in the course.

In addition to the mastery quizzes, you can also raise your mastery score by doing well on exams. Doing well on a particular topic on an exam will mean (in addition to getting a good score on the exam) that you get the equivalent of a 2/2 on a mastery quiz for that topic.

If this all seems like a rather convoluted way to do assessments, I just ask you to bear with me for the time being. **Once we do a few mastery quizzes I think the system will become completely transparent**, and if you still feel unsure how the system works or unclear on what your grade is at any point, please let me know! I'm always happy to go over it with you. Convoluted though it may be, this approach does have a purpose! The vast majority of the material is going to be *completely new*, and mastery grading allows you to be a little bit more flexible in how you absorb all this new information coming at you. For example, if in the middle of the course you reach a point where computing surface areas and arc lengths (S4) is absolutely no problem for you, but improper integrals (S3) still seem really opaque, you can spend your time focusing on improper integrals. As long as you eventually master arc length and surface area, it shouldn't impact your grade that it took a bit longer to master than some of the other topics.

Exams: There will be a midterm exam on July 21st and a final exam on August 7th. I'll post mock exams before each so you have an idea of what to expect, and at least before the final I'll hold some extra office hours—I'll try to do this before the midterms as well (TBA). If for some acceptable reason you cannot take the midterm, I will adjust the weight of the final exam to cover the midterm as well. (The following section gives an exception.)

Participation: I'll be posing questions frequently, and you'll help things along by answering them, even if you answer them incorrectly! In fact, incorrect answers can be especially helpful, and I'll sometimes ask for them explicitly when there's something we might really *like* to be able to do or something that really *seems* like it should work but for one reason or another doesn't. Thinking through why some things don't work is really just as important as seeing why other things do work.

Religious holidays and other excused absences

In accordance with University policy, students should notify faculty during the first week of the semester of their intention to be absent from class on their day(s) of religious observance. For details and policy, see Religious Holidays at the Office of the Provost at https://provost.gwu.edu/policies-procedures-and-guidelines.

Accommodations

Any student who may need an accommodation based on the impact of a disability should contact the Office of Disability Support Services (DSS) to inquire about the documentation necessary to establish eligibility and to coordinate a plan of reasonable and appropriate accommodations. DSS is located in Rome Hall, Suite 102. For additional information, please call DSS at 202-994-8250, or consult https://disabilitysupport.gwu.edu.

Academic integrity

The complete GWU academic integrity code can be viewed at https://studentconduct.gwu.edu. Per the code, "academic dishonesty is defined as cheating of any kind, including misrepresenting ones own work, taking credit for the work of others without crediting them and without appropriate authorization, and the fabrication of information." One thing to highlight here is that the definition given of cheating includes "engaging in unauthorized collaboration in any academic exercise." In this course there will be some assignments on which collaboration will not be allowed (e.g. Mastery Quizzes), and some on which it will be okay (e.g. the WeBWorK, although this is for your own practice, so if you do collaborate, make sure you understand everything your collaborators do). If you're ever unsure about whether collaboration is permitted on a particular assignment, please ask me.

Safety and security

In the case of an emergency, if at all possible, the class should shelter in place. If the building that the class is in is affected, follow the evacuation procedures for the building. After evacuation, seek shelter at a predetermined rendezvous location. Review the Emergency Response Handbook at https://safety.gwu.edu/emergency-response-handbook.

Schedule

This is an overview of the topics we'll be covering, along with when I think we'll cover things (but this is subject to change). Red cells below are exams and blue cells are when I anticipate having mastery quizzes due (usually these will be Tuesdays and Thursdays, with some exceptions at the end of the term).

June 30	Syllabus, inverse functions and derivatives	July 21	Midterm exam
July 1	Inverse trigonometric functions, exponentials and logarithms	July 22	Sequences
July 2	Derivatives and integrals with exponentials and logarithms	July 23	Series, divergence and integral tests
July 3	Derivatives and integrals with inverse trig	July 24	Comparison tests, alternating series test
July 7	L'Hôpital's rule	July 28	Ratio and root tests, power series
July 8	Integration by parts	July 29	Properties of power series
July 9	Trig integrals and trig substitutions	July 30	Taylor series
July 10	Partial fractions	July 31	Working with Taylor series
July 14	Improper integrals	Aug 4	Calculus with parametric equations
July 15	Arc length and surface area	Aug 5	Calculus with polar coordinates, area and arc length
July 16	Exponential growth, intro to differential equations	Aug 6	Review day
July 17	Separable equations	Aug 7	More on solids of revolution, final review

There will likely be an additional, optional mastery quiz due August 8th, after the final exam.